

Those questions highlighted in yellow are flagged out in the examiner's report as challenging.

1 You are given that $I = \int x(1-x)^{\frac{1}{2}} dx$.

(i) Use integration by parts to find an expression for I . [2]

(ii) Use the substitution $u^2 = 1-x$ to find another expression for I . [2]

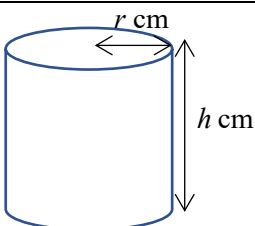
(iii) Show algebraically that your answers to parts (i) and (ii) differ by a constant. [2]

Qn	Solutions	Comments
1i	$I = \int x(1-x)^{\frac{1}{2}} dx$ $= -\frac{2}{3}x(1-x)^{\frac{3}{2}} + \frac{2}{3} \int (1-x)^{\frac{3}{2}} dx$ $= -\frac{2}{3}x(1-x)^{\frac{3}{2}} + \frac{2}{3}(1-x)^{\frac{5}{2}} \left(-\frac{2}{5}\right) + C_1$ $= -\frac{2}{3}x(1-x)^{\frac{3}{2}} - \frac{4}{15}(1-x)^{\frac{5}{2}} + C_1$	LIATE does not work here. Instead recall that in the product of 2 functions, there is one that will become a constant after repeated differentiation – choose it as u .
1ii	$I = \int x(1-x)^{\frac{1}{2}} dx$ $= \int (1-u^2)(u) (-2u) du$ $= \int (2u^4 - 2u^2) du$ $= \frac{2}{5}u^5 - \frac{2}{3}u^3 + C_2$ $= \frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + C_2$	$u^2 = 1-x \Rightarrow 2u \frac{du}{dx} = -1$
1iii	$-\frac{2}{3}x(1-x)^{\frac{3}{2}} - \frac{4}{15}(1-x)^{\frac{5}{2}} + C_1 - \left(\frac{2}{5}(1-x)^{\frac{5}{2}} - \frac{2}{3}(1-x)^{\frac{3}{2}} + C_2 \right)$ $= (1-x)^{\frac{3}{2}} \left[-\frac{2}{3}x - \frac{4}{15}(1-x) - \frac{2}{5}(1-x) + \frac{2}{3} \right] + C_1 - C_2$ $= (1-x)^{\frac{3}{2}} \left(\frac{-10x + 4x + 6x - 4 - 6 + 10}{15} \right) + C_1 - C_2$ $= (1-x)^{\frac{3}{2}} (0) + C_1 - C_2$ $= C_1 - C_2, \text{ a constant}$ Answers to parts (i) and (ii) differ by a constant.	Though the question asks for difference, there is no need to consider the absolute value since we only need to show the difference is a constant, which can be positive or negative. This is a “show” question, hence must include relevant details. Take note of the algebraic manipulations.

- 2 (i) Sketch the graph of $y = \frac{2-x}{3x^2+5x-8}$. Give the equations of the asymptotes and the coordinates of the point(s) where the curve crosses either axis. [4]
- (ii) Solve the inequality $\frac{2-x}{3x^2+5x-8} > 0$. [1]
- (iii) Hence solve the inequality $\frac{x-2}{3x^2+5x-8} > 0$. [1]

Qn	Solutions	Comments
2i	$y = \frac{2-x}{3x^2+5x-8} = \frac{2-x}{(3x+8)(x-1)}$ Asymptotes: $x = -\frac{8}{3}$, $x = 1$ and $y = 0$ Axial Intercepts: $(0, -\frac{1}{4})$ and $(2, 0)$	Always refer to the equation of the curve when finding the following • equations of asymptotes • axial intercepts Use the features found to guide you. Do not just rely on GC for the shape of the curve. Change window settings for x_{min} to be 1, x_{max} to be 6, then select zoomfit to see the detail intercept at $(2, 0)$, you should realise that the curve will go down below the x-axis when $x > 2$. Investigate further by changing window settings or use zoom function in GC.
2ii	From the graph, $\frac{2-x}{3x^2+5x-8} > 0 \Rightarrow x < -\frac{8}{3} \text{ or } 1 < x < 2$	Use $\frac{2-x}{(3x+8)(x-1)}$ to check.
2iii	$\frac{x-2}{3x^2+5x-8} > 0 \Rightarrow \frac{2-x}{3x^2+5x-8} < 0$ $\Rightarrow -\frac{8}{3} < x < 1 \text{ or } x > 2$	

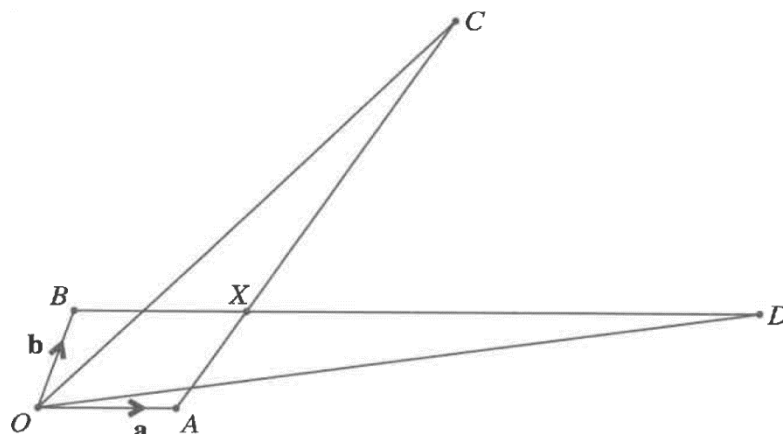
- 3 A solid cylinder has radius r cm, height h cm and total surface area 900 cm^2 . Find the exact value of the maximum possible volume of the cylinder. Find also the ratio $r : h$ that gives this maximum volume. [7]

Qn	Solutions	Comments
3	Total surface area = 900 cm^2 $2\pi r^2 + 2\pi rh = 900$ $h = \frac{450 - \pi r^2}{\pi r}$ Volume of cylinder, $V = \pi r^2 h = \pi r^2 \left(\frac{450 - \pi r^2}{\pi r} \right)$ $\therefore V = 450r - \pi r^3$ $\frac{dV}{dr} = 450 - 3\pi r^2$ $\frac{dV}{dr} = 0 \Rightarrow \pi r^2 = 150$ $\therefore r = \sqrt{\frac{150}{\pi}}$ $\frac{d^2V}{dr^2} = -6\pi r < 0 \text{ when } r = \sqrt{\frac{150}{\pi}} > 0.$ So V is maximum when $r = \sqrt{\frac{150}{\pi}}$. Exact value of the maximum possible volume of cylinder $= 450\sqrt{\frac{150}{\pi}} - \pi \left(\sqrt{\frac{150}{\pi}} \right)^3$ $= 450\sqrt{\frac{150}{\pi}} - 150\sqrt{\frac{150}{\pi}}$ $= 300\sqrt{\frac{150}{\pi}} = 1500\sqrt{\frac{6}{\pi}} \text{ cm}^3$ $r = \sqrt{\frac{150}{\pi}}, h = \frac{450 - 150}{\pi \left(\sqrt{\frac{150}{\pi}} \right)} = \frac{300}{\sqrt{150\pi}}$ $\frac{r}{h} = \sqrt{\frac{150}{\pi}} \times \frac{\sqrt{150\pi}}{300} = \frac{150}{300} = \frac{1}{2}$ $\therefore r : h = 1 : 2$	 We need exact value. Don't forget to perform derivative test to show maximum. Use second derivative test since it is easy to differentiate $\frac{dV}{dr}$ Answer to leave as a single simplified expression Again, need to be careful with your algebraic manipulation.

- 4 (i) Given that $f(x) = \sec 2x$, find $f'(x)$ and $f''(x)$. Hence, or otherwise, find the Maclaurin series for $f(x)$, up to and including the term in x^2 . [5]
- (ii) Use your series from part (i) to estimate $\int_0^{0.02} \sec 2x \, dx$, correct to 5 decimal places. [2]
- (iii) Use your calculator to find $\int_0^{0.02} \sec 2x \, dx$, correct to 5 decimal places. [1]
- (iv) Comparing your answers to parts (ii) and (iii), and with reference to the value of x , comment on the accuracy of your approximations. [2]
- (v) Explain why a Maclaurin series for $g(x) = \operatorname{cosec} 2x$ cannot be found. [1]

Qn	Solutions	Comments
4i	$f(x) = \sec 2x$ $f'(x) = 2 \sec 2x \tan 2x$ $f''(x) = 2 \{ (\sec 2x)(2 \sec^2 2x) + (\tan 2x)(2 \sec 2x \tan 2x) \}$ $= 4 \sec^3 2x + 4 \tan^2 2x \sec 2x$ When $x = 0$, $f(0) = 1$, $f'(0) = 2(1)(0) = 0$ $f''(0) = 4(1) + 4(0)(1) = 4$ $f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots$ $= 1 + x(0) + \frac{4}{2!} x^2 + \dots$ $= 1 + 2x^2 + \dots$	
4ii	$\int_0^{0.02} \sec 2x \, dx \approx \int_0^{0.02} (1 + 2x^2) \, dx = 0.02001$ (to 5 d.p.)	
4iii	$\int_0^{0.02} \sec 2x \, dx = 0.02001$ (to 5 d.p.)	
4iv	Answers to (ii) and (iii) are equal to 5 decimal places. Since $0 \leq x \leq 0.02$ is small, x^3 and terms with higher power can be ignored. $\therefore \sec 2x \approx 1 + 2x^2$ So the approximation is accurate for small x .	
4v	$g(x) = \operatorname{cosec} 2x = \frac{1}{\sin 2x}$ When $x = 0$, $g(0)$ is undefined since $\sin 2x = 0$. So a Maclaurin series for $g(x)$ cannot be found.	

5



With reference to the origin O , the points A , B , C and D are such that $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = 2\mathbf{a} + 4\mathbf{b}$ and $\overrightarrow{OD} = \mathbf{b} + 5\mathbf{a}$. The lines BD and AC cross at X (see diagram).

(i) Express $\overrightarrow{OX}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. [4]

The point Y lies on CD and is such that the points O , X and Y are collinear.

(ii) Express $\overrightarrow{OY}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ and find the ratio $OX : OY$. [6]

Qn	Solutions	Comments
5i	$\overrightarrow{AC} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a} = \mathbf{a} + 4\mathbf{b}$ $\overrightarrow{BD} = \mathbf{b} + 5\mathbf{a} - \mathbf{b} = 5\mathbf{a}$ Equation of line AC : $\mathbf{r} = \mathbf{a} + \lambda(\mathbf{a} + 4\mathbf{b})$, $\lambda \in \mathbb{R}$ Equation of line BD : $\mathbf{r} = \mathbf{b} + \mu(5\mathbf{a})$, $\mu \in \mathbb{R}$ Since X is the point of intersection of lines BD and AC $\mathbf{a} + \lambda(\mathbf{a} + 4\mathbf{b}) = \mathbf{b} + \mu(5\mathbf{a})$ $(1 + \lambda - 5\mu)\mathbf{a} + (4\lambda - 1)\mathbf{b} = \mathbf{0}$ $(1 + \lambda - 5\mu)\mathbf{a} = (1 - 4\lambda)\mathbf{b}$ Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors, $\therefore 1 - 4\lambda = 0 \Rightarrow \lambda = \frac{1}{4}$ and $1 + \lambda - 5\mu = 0 \Rightarrow \mu = \frac{1}{5}\left(1 + \frac{1}{4}\right) = \frac{1}{4}$ $\overrightarrow{OX} = \frac{5}{4}\mathbf{a} + \mathbf{b}$	
5ii	$\overrightarrow{CD} = \mathbf{b} + 5\mathbf{a} - 2\mathbf{a} - 4\mathbf{b} = 3\mathbf{a} - 3\mathbf{b}$ Equation of line CD : $\mathbf{r} = 2\mathbf{a} + 4\mathbf{b} + k(\mathbf{a} - \mathbf{b})$, $k \in \mathbb{R}$ Y lies on CD , $\overrightarrow{OY} = 2\mathbf{a} + 4\mathbf{b} + k(\mathbf{a} - \mathbf{b})$ $= (2 + k)\mathbf{a} + (4 - k)\mathbf{b}$ O , X , and Y are collinear, $\overrightarrow{OY} = t\overrightarrow{OX} = t\left(\frac{5}{4}\mathbf{a} + \mathbf{b}\right)$	

Qn	Solutions	Comments
	$(2+k)\mathbf{a} + (4-k)\mathbf{b} = t\left(\frac{5}{4}\mathbf{a} + \mathbf{b}\right)$ $\left(2+k-\frac{5}{4}t\right)\mathbf{a} = (-4+k+t)\mathbf{b}$ Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors, $\left. \begin{aligned} 2+k-\frac{5}{4}t &= 0 \\ -4+k+t &= 0 \end{aligned} \right\}$ Using GC, $k = \frac{4}{3}$, $t = \frac{8}{3}$ $\overrightarrow{OY} = \frac{8}{3}\overrightarrow{OX} \Rightarrow \frac{8}{3}\left(\frac{5}{4}\mathbf{a} + \mathbf{b}\right) = \frac{10}{3}\mathbf{a} + \frac{8}{3}\mathbf{b}$ $\overrightarrow{OY} = \frac{8}{3}\overrightarrow{OX} \Rightarrow \frac{ \overrightarrow{OX} }{ \overrightarrow{OY} } = \frac{3}{8}$ $\therefore OX : OY = 3 : 8$	Use: PlySmlt2 to solve the simultaneous equations.

- 6 In a certain country there are 100 professional football clubs, arranged in 4 divisions. There are 22 clubs in Division One, 24 in Division Two, 26 in Division Three and 28 in Division Four.

- (i) Alice wishes to find out about approaches to training by clubs in Division One, so she sends a questionnaire to the 22 clubs in Division One. Explain whether these 22 clubs form a sample or a population. [1]
- (ii) Dilip wishes to investigate the facilities for supporters at the football clubs, but does not want to obtain the detailed information necessary from all 100 clubs. Explain how he should carry out his investigation, and why he should do the investigation in this way. [2]
- (iii) Find the number of different possible samples of 20 football clubs, with 5 clubs chosen from each division. [3]

Qn	Solutions	Comments
6i	Alice is only interested in the approaches to training by clubs in Division One . Since there are only 22 clubs in Division One in the country and she sends questionnaire to all the 22 clubs, these 22 clubs form a population.	The intent of the experiment determines whether a set is a population or sample. "Explain" requires us to justify our answer. We need to make it clear that there are only 22 Division One clubs in the country.
6ii	Dilip should take a random sample of the clubs so that he can avoid bias. He should select the sample in such a way that every club has an equal chance of being selected and the selections are independent of each other.	
6iii	Number of different possible samples of 20 football clubs $= {}^{22}C_5 \times {}^{24}C_5 \times {}^{26}C_5 \times {}^{28}C_5$ $= 7.2361 \times 10^{18}$ $= 7.24 \times 10^{18}$	

- 7 A company produces drinking mugs. It is known that, on average, 8% of the mugs are faulty. Each day the quality manager collects 50 of the mugs at random and checks them; the number of faulty mugs found is the random variable F .

- (i) State, in the context of the question, two assumptions needed to model F by a binomial distribution. [2]

You are now given that F can be modelled by a binomial distribution.

- (ii) Find the probability that, on a randomly chosen day, at least 7 faulty mugs are found. [2]
- (iii) The number of faulty mugs produced each day is independent of other days. Find the probability that, in a randomly chosen working week of 5 days, at least 7 faulty mugs are found on no more than 2 days. [2]

The company also makes saucers. The number of faulty saucers also follows a binomial distribution. The probability that a saucer is faulty is p . Faults on saucers are independent of faults on mugs.

- (iv) Write down an expression in terms of p for the probability that, in a random sample of 10 saucers, exactly 2 are faulty. [1]

The mugs and saucers are sold in sets of 2 randomly chosen mugs and 2 randomly chosen saucers. The probability that a set contains at most 1 faulty item is 0.97.

- (v) Write down an equation satisfied by p . Hence find the value of p . [4]

Qn	Solutions	Comments
7i	(1) The probability of a mug being faulty remains at 0.08. (2) A mug being faulty is independent of any other mug being faulty.	Check your phrasing. Wrong answers: - Probability of a mug being faulty is independent ... The following are incorrect assumptions - Number of mugs collected is fixed - A mug is either faulty or not. These conditions are required for a binomial distribution, but they are covered in the description given in the question, so they are not assumptions that need to be made.
7ii	$F \sim B(50, 0.08)$ $P(F \geq 7) = 1 - P(F \leq 6)$ $= 0.10187 \approx 0.102$	
7iii	Let Y be the number of days on which at least 7 faulty mugs are found in a working week of 5 days. $Y \sim B(5, 0.10187)$ $P(Y \leq 2) = 0.99098 \approx 0.991$	Use at least 5sf in the answer from (ii).
7iv	$P(\text{exactly 2 faulty saucers in a sample of 10})$	Alternatively, we can also solve this question using $B(10, p)$.

Qn	Solutions	Comments
	$= {}^{10}C_2 p^2 (1-p)^8$ $= 45 p^2 (1-p)^8$	
7v	P(a set contains at most 1 faulty item) = 0.97 P(no faulty items) + P(exactly 1 mug is faulty) + P(exactly 1 saucer is faulty) = 0.97 $0.92^2 (1-p)^2 + {}^2C_1 (0.08)(0.92)(1-p)^2$ $+ {}^2C_1 p(1-p)(0.92)^2 = 0.97$ $0.8464(1-2p+p^2) + 0.1472(1-2p+p^2)$ $+ 1.6928(p-p^2) = 0.97$ $-0.6992p^2 - 0.2944p + 0.9936 = 0.97$ $0.6992p^2 + 0.2944p - 0.0236 = 0$ $\therefore p = 0.068891 \approx 0.0689$	A set consists of 2 mugs and 2 saucers “at most 1 faulty item” implies No faulty item OR exactly 1 faulty item. Exactly 1 faulty saucer OR exactly 1 faulty mug 2C_1: Choose one of the two mugs to be faulty 2C_1: Choose one of the two saucers to be faulty

- 8** Gerri collects characters given away in packets of breakfast cereal. There are four different characters: Horse, Rider, Dog and Bird. Each character is made in four different colours: Orange, Yellow, Green and White. Gerri has collected 56 items; the numbers of each character and colour are shown in the table.

	Orange	Yellow	Green	White
Horse	1	1	3	4
Rider	1	1	7	5
Dog	3	7	1	6
Bird	4	5	6	1

- (i) Gerri puts all the items in a bag and chooses one item at random.
- (a) Find the probability that this item is either a Horse or a Rider. [1]
- (b) Find the probability that this item is either a Dog or a Bird but the item is not White. [1]
- (ii) Gerri now puts the item back in the bag and chooses two items at random.
- (a) Find the probability that both of the items are Horses, but neither of the items is Orange. [1]
- (b) Find the probability that Gerri's two items include exactly one Dog and exactly one item that is Yellow. [3]
- (iii) Gerri has two favourites among the 16 possible colour/character combinations. The probability of choosing these two at random from the 56 items is $\frac{1}{77}$. Write down all the possibilities for Gerri's two favourite colour/character combinations. [3]

Qn	Solutions	Comments
8ia	$P(\text{item is either a Horse or a Rider})$ $= \frac{9}{56} + \frac{14}{56} = \frac{23}{56}$ (or 0.411)	Secondary school work on probability
8ib	$P(\text{item is either a Dog or a Bird but the item is not white})$ $= \frac{11}{56} + \frac{15}{56} = \frac{26}{56} = \frac{13}{28}$ (or 0.464)	Secondary school work on probability
8iia	$P(\text{both items are Horses but neither of the items is orange})$ $= \frac{8}{56} \times \frac{7}{55} = \frac{1}{55}$ (0.0182)	No. of non-orange horses = $1 + 3 + 4 = 8$ "chooses two items at random" is the same as taking items out 1 by 1 without replacement.
8iib	$P(\text{exactly 1 item is Dog and exactly 1 item is Yellow})$ $= P(\text{Dog Yellow \& the other item is not Dog \& not Yellow}) + P(\text{Dog not Yellow \& the other item is Yellow \& not Dog})$ $= \frac{7}{56} \times \frac{56-17-7}{55} \times 2 + \frac{10}{56} \times \frac{7}{55} \times 2$ $= \frac{8}{55} + \frac{1}{22} = \frac{21}{110}$	
8iii	Let x and y be the 2 cell entries of Gerri's two favourite colour/character combinations $P(\text{these 2 are chosen at random from the 56 items})$ $= \frac{x}{56} \times \frac{y}{55} \times 2 = \frac{xy}{28 \times 55}$ $\frac{xy}{28 \times 55} = \frac{1}{77} \Rightarrow xy = 20$ From the cell entries, $xy = 20 \Rightarrow x = 4$ and $y = 5$ or $x = 5$ and $y = 4$. Hence Gerri's favourite combinations are: (1) White Horse and White Rider (2) White Horse and Yellow Bird (3) Orange Bird and White Rider (4) Orange Bird and Yellow Bird	

- 9 A company produces resistors rated at 750 ohms for use in electronic circuits. The production manager wishes to test whether the mean resistance of these resistors is in fact 750 ohms. He knows that the resistances are normally distributed with variance 100 ohms².

(i) Explain whether the manager should carry out a 1-tail test or a 2-tail test. State hypotheses for the test, **defining any symbols you use.** [2]

The production manager takes a random sample of 8 of these resistors. He finds that the resistances in ohms, are as follows.

742 771 768 738 769 752 742 766

(ii) Find the mean of the sample of 8 resistors. Carry out the test, at the 5% level of significance, for the production manager. Give your conclusion in context. [5]

The company also produces resistors rated at 1250 ohms. Nothing is known about the distribution of the resistances of these resistors.

(iii) Describe how, and why, a test of the mean resistance of the 1250 ohms resistors would need to differ from that for the 750 ohms resistors. [2]

Qn	Solutions	Comments
9i	A 2-tail test should be done since the manager wishes to test whether the mean resistance of the resistors differs from 750 ohms. Let μ ohms be the population mean resistance of the resistors rated at 750 ohms. Null hypothesis, $H_0: \mu = 750$ Alternative hypothesis, $H_1: \mu \neq 750$	“Explain” means you need to justify your answer. Need to define any symbols you used i.e. define μ
9ii	Let X ohms be the resistance of a randomly selected resistor rated at 750 ohms. Level of significance: 5% Test statistic: When H_0 is true, $Z = \frac{\bar{X} - 750}{\frac{10}{\sqrt{n}}} \sim N(0,1)$ Computation: $n = 8$, $\bar{x} = \frac{\sum x}{n} = \frac{6048}{8} = 756$ $p\text{-value} = 0.089686 \approx 0.0897$ Conclusion: Since $p\text{-value} = 0.0897 > 0.05$, H_0 is not rejected at 5% level of significance. Hence there is insufficient evidence to conclude that the population mean resistance of the resistors is not 750 ohms.	
9iii	Since nothing is known about the distribution of the resistances of the resistors rated as 1250 ohms, a large random sample of size n (at least 30) must be taken so that by Central Limit	

Qn	Solutions	Comments
	Theorem, the sample mean resistance ($\bar{X}$) of the resistors rated at 1250 ohms is approximately normally distributed. We also need to estimate the population variance using the unbiased estimator, S^2 . It is a good estimator of the population variance, σ^2 when n is large.	

- 10** Abi and Bhani find the fuel consumption for a car driven at different constant speeds. The table shows the fuel consumption, y kilometres per litre, for different constant speeds, x kilometres per hour.

x	40	45	50	55	60
y	22	20	18	17	16

- (i)** Abi decides to model the data using the line $y = 35 - \frac{1}{3}x$.

- (a)** On the grid opposite

- draw a scatter diagram of the data,
- draw the line $y = 35 - \frac{1}{3}x$. [2]

- (b)** For a line of best fit $y = f(x)$, the residual for a point (a, b) plotted on the scatter diagram is the vertical distance between $(a, f(a))$ and (a, b) . Mark the residual for each point on your diagram. [1]

- (c)** Calculate the sum of the squares of the residuals for Abi's line. [1]

- (d)** Explain why, in general, the sum of the squares of the residuals rather than the sum of the residuals is used. [1]

Bhani models the same data using a straight line passing through the points (40, 22) and (55, 17). The sum of the squares of the residuals for Bhani's line is 1.

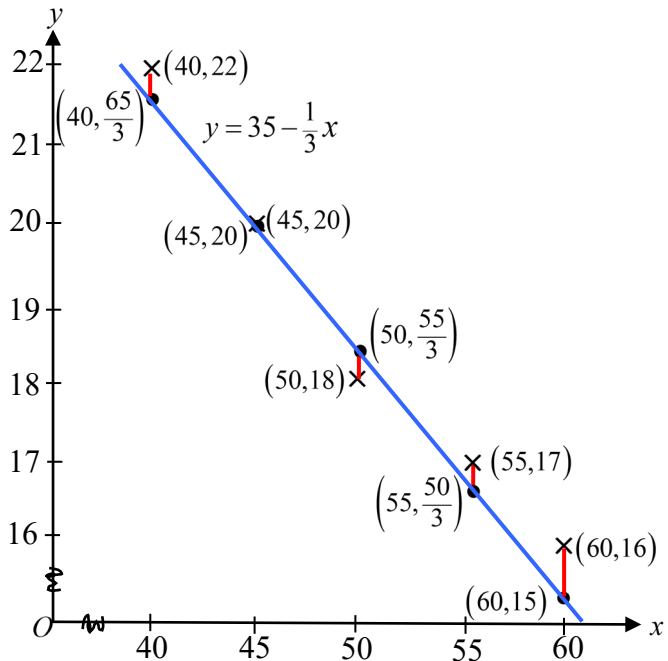
- (ii)** State, with a reason, which of the two models, Abi's or Bhani's, gives a better fit. [1]

- (iii)** State the coordinates of the point that the least squares regression line must pass through. [1]

- (iv)** Use your calculator to find the equation of the least squares regression line of y on x . State the value of the product moment correlation coefficient. [3]

- (v)** Use the equation of the regression line to estimate the fuel consumption when the speed is 30 kilometres per hour. Explain whether you would expect this value to be reliable. [2]

- (vi)** Cerie performs a similar experiment on a different car. She finds that the sum of the squares of the residuals for her line is 0. What can you deduce about the data points in Cerie's experiment? [1]

Qn	Solutions	Comments
10ia 10ib	 A scatter plot on a Cartesian coordinate system. The x-axis ranges from 40 to 60 with major ticks every 5 units. The y-axis ranges from 16 to 22 with major ticks every 1 unit. A blue line of best fit is drawn with the equation $y = 35 - \frac{1}{3}x$. Six data points are plotted as black dots with their coordinates labeled: $(40, \frac{65}{3})$, $(45, 20)$, $(50, \frac{55}{3})$, $(55, \frac{50}{3})$, $(60, 15)$, and $(60, 16)$. Red vertical line segments connect each point to the blue line, representing the residuals. The points $(40, 22)$, $(45, 20)$, $(50, 18)$, $(55, 17)$, and $(60, 16)$ are marked with red 'x' symbols on the line.	The residual for a point (a, b) is the vertical distance between $(a, f(a))$ and (a, b) hence the residuals should be drawn vertically.
10ic	Sum of the residuals for Abi's line $= \left(22 - \frac{65}{3}\right)^2 + (20 - 20)^2 + \left(18 - \frac{55}{3}\right)^2 + \left(17 - \frac{50}{3}\right)^2 + (16 - 15)^2$ $= \left(\frac{1}{3}\right)^2 + 0^2 + \left(-\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + 1^2 = \frac{4}{3}$	i.e. sum of the squares of the vertical distances
10id	The residuals of the points on the scatter diagram are either positive, negative or zero. If we take sum of the residuals, the positive and negative values will cancel off and will not give a good indication of how much the points deviate from the line. Hence the sum of the squares of the residuals is used so that the contribution of all the residuals will be non-negative.	
10ii	Sum of squares for Abi's line $= \frac{4}{3} > 1$ which is the sum of squares of Bhani's line, this indicates that the points in the scatter diagram lie closer to Bhani's line. So Bhani's model is a better fit.	
10iii	The least squares regression line passes through the point $(\bar{x}, \bar{y})$ i.e. $(50, 18.6)$.	
10iv	Equation of the least squares regression line of y on x is given by $y = 33.6 - 0.3x$ The product moment correlation coefficient is $r = -0.98479 \approx -0.985$	
10v	When $x = 30$, $y = 33.6 - 0.3(30) = 24.6$. The fuel consumption is 24.6 km/litre.	

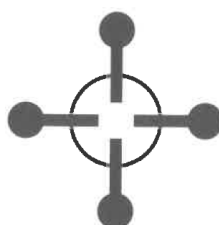
Qn	Solutions	Comments
	Since $x = 30$ lies outside the data range of $40 \leq x \leq 60$, so extrapolation is used. Hence the value 24.6 is not reliable.	
10vi	The sum of the squares of the residuals for Cerie's line is 0 means that all the points on the scatter diagram lie on her line.	

11 In this question you should state clearly all the distributions that you use, together with the values of the appropriate parameters.

Arif is making models of hydrocarbon molecules. Hydrocarbons are chemical compounds made from carbon atoms and hydrogen atoms. Arif has a bag containing a large number of white balls to represent the carbon atoms, and a bag containing a large number of black balls to represent the hydrogen atoms. The masses of the white balls have the distribution $N(110, 4^2)$ and the masses of the black balls have the distribution $N(55, 2^2)$. The units for mass are grams.

- (i) Find the probability that the total mass of 4 randomly chosen white balls is more than 425 grams. [2]
- (ii) Find the probability that the total mass of a randomly chosen white ball and a randomly chosen black ball is between 161 and 175 grams. [2]
- (iii) The probability that 2 randomly chosen white balls and 3 randomly chosen black balls have total mass less than M grams is 0.271. Find the value of M . [4]

Arif also has a bag containing a large number of connecting rods to fix the balls together. The masses of the connecting rods, in grams, have the distribution $N(20, 0.9^2)$. In order to make models of methane (a hydrocarbon), Arif has to drill 1 hole in each black ball, and 4 holes in each white ball, for the connecting rods to fit in. This reduces the mass of each black ball by 10% and reduces the mass of each white ball by 30%.



A methane molecule consists of 1 carbon atom and 4 hydrogen atoms. Arif makes a model of a methane molecule using 4 black balls, 1 white ball and 4 connecting rods (see diagram). The balls and connecting rods are all chosen at random.

- (iv) Find the probability that the mass of Arif's model is more than 350 grams. [4]

Qn	Solutions	Comments
11i	Let X grams be the mass of a white ball and Y grams be the mass of a black ball. $X \sim N(110, 4^2)$, $Y \sim N(55, 2^2)$ Let $U = X_1 + X_2 + X_3 + X_4$ be the total mass of the chosen 4 white balls.	

Qn	Solutions	Comments
	$E(U) = 4E(X) = 4 \times 110 = 440$ $\text{Var}(U) = 4\text{Var}(X) = 4 \times 4^2 = 8^2$ $U \sim N(440, 8^2)$ $P(U > 425) = 0.96960 \approx 0.970$	
11ii	Let $V = X + Y$ be the total mass of a white ball and a black ball. $E(V) = E(X) + E(Y) = 110 + 55 = 165$ $\text{Var}(V) = \text{Var}(X) + \text{Var}(Y) = 4^2 + 2^2 = 20$ $V \sim N(165, 20)$ $P(161 < V < 175) = 0.80178 \approx 0.802$	
11iii	Let $W = X_1 + X_2 + Y_1 + Y_2 + Y_3$ be the mass of the 2 white balls and the 3 black balls. $E(W) = 2E(X) + 3E(Y) = 2 \times 110 + 3 \times 55 = 385$ $\text{Var}(W) = 2\text{Var}(X) + 3\text{Var}(Y) = 2 \times 4^2 + 3 \times 2^2 = 44$ $W \sim N(385, 44)$ Given $P(W < M) = 0.271$ From GC, $P(W < 380.96) \approx 0.271$ $\therefore M = 381$ (3 s.f.)	Correct to nearest 3 s.f, not round down.
11iv	Let C grams be the mass of a connecting rod. $C \sim N(20, 0.9^2)$ Let T grams be the mass of Arif's model. $T = C_1 + C_2 + C_3 + C_4 + 0.7X + 0.9(Y_1 + Y_2 + Y_3 + Y_4)$ $E(T) = 4E(C) + 0.7E(X) + 0.9 \times 4E(Y)$ $= 4 \times 20 + 0.7 \times 110 + 0.9 \times 4 \times 55 = 355$ $\text{Var}(T) = 4\text{Var}(C) + 0.7^2 \text{Var}(X) + 0.9^2 \times 4 \times \text{Var}(Y)$ $= 4 \times 0.9^2 + 0.7^2 \times 4^2 + 0.9^2 \times 4 \times 2^2 = 24.04$ $T \sim N(355, 24.04)$ $P(T > 350) = 0.84608 \approx 0.846$	$\text{Var}(0.9Y) = 0.9^2 \text{Var}(Y)$